FEAT-ZD-48600 · FE-03 Identify the components/layers to be changed · Components, layers, sources or pipelines to change

Frontend agent · item task.components · source hobasa-frontend-agent/outputs/FEAT-ZD-48600/task.components.md · approved by automated checks and independent judge

task.components: FEAT-ZD-48600 (Frontend lane)

This file lists the components and layers of `ap_portal` that the Frontend task must create or update, and how to do it with the fewest lines of code. Step: FE-03, Identify the components/layers to be changed.

Statements marked **agent judgement** are this agent's own reading of the code and inputs; no input states them as fact. This step changes no code. FE-04 does the coding.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	FE-03 Identify the components/layers to be changed	record:brief
Owner / gate	Developer / automated checks, independent judge, sampled human evaluation	record:brief
Task in scope	TSK-07 (STORY-07): render the overdue risk table SC-1 with `renderOverdueRisk`	feat.tasks (sections 1 and 3, TSK-07); task.repo_access (header)
Design component	HC-8 `renderOverdueRisk` (new); HC-9 `formatAmount`, `renderAgingSummary` (unchanged)	feat.design (A2, B4.1)
Repository and branch	`ap_portal`, branch `feature/story` at `ff761ee`, write access confirmed	task.repo_access (section 1)
Inputs read	feat.tasks, feat.stories, feat.prd, feat.trd, feat.design, task.repo_access	record:brief
Code read	`src/ap_portal/index.js`, `tests/portal.test.js`, `package.json`, `ci.mjs`, `README.md`, `CLAUDE.md`	tool:repo; file:ap_portal/package.json; file:ap_portal/ci.mjs; file:ap_portal/README.md; file:ap_portal/CLAUDE.md

1. Result

One source module, one test file and one version field change. No new file, no new layer and no new dependency is needed.

#	Component / layer	File	Change	Source
1	View helper layer (the only runtime layer in ap_portal)	`src/ap_portal/index.js`	Update, additive: two constants, three private helpers, one exported function `renderOverdueRisk`, the `VERSION` step and one more name in `module.exports`; optionally the module header comment (agent judgement, item 1.1)	feat.design (B4.1, B4.2); feat.tasks (TSK-07); tool:repo

#	Component / layer	File	Change	Source
2	Unit tests	`tests/portal.test.js`	Update, additive: new `node:test` tests appended; the two existing tests unchanged	feat.design (B4.5); feat.tasks (TSK-07); tool:repo
3	Package metadata	`package.json`	Update: `"version"` only, moved together with `VERSION`	feat.tasks (section 2, TSK-07); file:ap_portal/package.json

Everything else in the repository stays unchanged (section 4).

2. What exists today (code read)

Item	Today	Source		
Runtime	Node.js 22, CommonJS; tests with `node:test`	file:ap_portal/CLAUDE.md; task.repo_access (section 2)		
Code and test folders	Code in `src/ap_portal/`, tests in `tests/`; version kept in `src/ap_portal/index.js`	file:ap_portal/CLAUDE.md		
Source files	One: `src/ap_portal/index.js`	task.repo_access (section 2); tool:repo		
Module contents	`'use strict'`; header comment "AP portal view helpers: render the aging summary returned by the ap_aging service as HTML."; `VERSION = '2.3.0'`; `LABELS`; `formatAmount(value)` (`Number(value \	\	0).toFixed(2)); render AgingSummary(buckets) (`template literal per row, rows.join(''), wrapped in <table class="aging-summary">`)	tool:repo
Exports	`{ VERSION, LABELS, formatAmount, renderAgingSummary }`	tool:repo		
Tests	`tests/portal.test.js`: `formats amounts with two decimals`; `renders every bucket in order`; the package is loaded with `require('ap_portal')`	tool:repo		
Package version	`"version": "2.3.0"` in `package.json`	file:ap_portal/package.json		
CI	`node ci.mjs`: `node --check` on every `.js` file in `src/ap_portal`; tests with `--experimental-test-coverage --test-coverage-include=src/**` and `NODE_PATH=src`; static checks fail a source line matching `console.log` or `eval(` or longer than 140 characters; passes only with coverage >= 90, no failed test and at least one test run	file:ap_portal/ci.mjs		

Item	Today	Source		
Layers not present	No transport or API client, no host page or routing, no stylesheet, no state store. `renderAgingSummary` takes a plain object and returns an HTML string	tool:repo; feat.design (A3 "Transport for CT-1", A1 UI boundary)		

Two findings from the code shape the change:

- `formatAmount` formats through a binary float, which conflicts with D-2026-044. The new code must not call it; the existing helper stays unchanged (feat.trd F-12, TQ-2; record:memory D-2026-044).
- `renderAgingSummary` escapes nothing. Its labels are constants, but the new table renders feed text (`id`, `vendor`), so that pattern must not be copied as it is (feat.trd F-13, R-8).

3. Components to create or update, with the minimum-lines approach

3.1 `src/ap_portal/index.js` *(view helper layer)*

Everything below goes into the existing module, next to the existing code, as feat.design B4.1 places it (DD-2). New items are added after `LABELS` and after `renderAgingSummary`; no existing line is rewritten except `VERSION`, the `module.exports` line and, optionally, the header comment.

#	Item	Kind	Exported	What it does	Minimum-lines approach (agent judgement unless a source says otherwise)	Source
1.1	Header comment (line 2)	Update (agent judgement; optional)	—	Mention the overdue risk table next to the aging summary, so the module comment stays accurate	Edit the one existing comment line; no new doc block. Neither feat.design nor feat.tasks lists this edit; it is proposed for the Developer to accept or drop	tool:repo
1.2	`VERSION`	Update	Yes (already)	Minor version step from `2.3.0`; the lane confirms the next number	Change the string literal only	feat.tasks (section 2, TSK-07); feat.trd (T-A6)
1.3	`RISK_CLASSES`	New constant	No	Allow-list `['high', 'medium', 'low', 'unknown']`	One line	feat.design (B4.2, B4.3)
1.4	`OVERDUE_HEADINGS`	New constant	No	`['Invoice', 'Vendor', 'Days overdue', 'Amount']` in order	One line; the header row is built from it with `map` + `join('')`, as `renderAgingSummary` builds its rows	feat.design (B4.2); feat.prd (AC-19); tool:repo

#	Item	Kind	Exported	What it does	Minimum-lines approach (agent judgement unless a source says otherwise)	Source
1.5	`escapeHtml(value)`	New private helper	No	Escapes &, <, >, ", '	A single `String(value).replace(/[&<>"']/g, ...)` with a small lookup object	feat.design (B4.2, SE-2); feat.trd (IF-7.7)
1.6	`toTwoDecimalString(value)`	New private helper	No	String-only amount: must be a string matching `^-?\d+(\.\d{1,2})?$`, padded to two decimals as a string (`"980.5"` → `980.50`); otherwise `TypeError`. Never `Number()`, `parseFloat` or `formatAmount`	One type check plus one regex match with capture groups; pad the fraction with `padEnd(2, '0')`. No numeric conversion at any point	feat.design (B4.2, B4.4, DD-5, DD-8); feat.trd (IF-7.5); feat.tasks (TSK-07 implementation notes); record:memory (D-2026-044, L-011)
1.7	`riskClass(risk)`	New private helper	No	`risk-<risk>` for a known value, else `risk-unknown`	One expression with `RISK_CLASSES.includes(risk)`	feat.design (B4.2); feat.trd (IF-7.4)
1.8	`renderOverdueRisk(rows)`	New function	Yes	`''` for `undefined`, `null` or `[]`; `TypeError` for any other non-array; otherwise `<table class="overdue-risk">` + plain header row (no class, no `<thead>`/`<tbody>`) + one `<tr class="risk-…">` per row in the given order with cells id, vendor, days, amount (`class="amount"`) + `</table>`; `TypeError` when `days` is not an integer	Same style as `renderAgingSummary`: template literals, `rows.map(...).join('')`. The `days` check is an inline `Number.isInteger` test inside the row mapper, not another helper. No sorting, no risk text, no Risk column, no row actions	feat.design (B4.2 algorithm steps 1 to 4, B4.4); feat.trd (IF-7.1 to IF-7.8); feat.tasks (TSK-07); feat.stories (AC-07.1 to AC-07.9)
1.9	`module.exports`	Update	—	Add `renderOverdueRisk` after the existing four names	Append one name to the existing line; the helpers and constants stay private, so the public surface grows by one function only	feat.design (B4.2); feat.stories (AC-07.10); file: ap_portal/CLAUDE.md ("Keep existing public signatures; add new functions for new behaviour.")

#	Item	Kind	Exported	What it does	Minimum-lines approach (agent judgement unless a source says otherwise)	Source
—	`LABELS`, `formatAmount`, `renderAgingSummary`	Unchanged	Yes (already)	—	Not touched and not reused by the new code	feat.design (HC-9, B4.1); feat.tasks (TSK-07 "Must not change"); feat.trd (C-9)

Why no new source file (agent judgement). feat.design B4.1 and feat.tasks TSK-07 put all of this in `src/ap_portal/index.js`; a new module would be outside the task's file budget. CI also compiles every `.js` file in `src/ap_portal` and measures coverage over `src/**` (file:ap_portal/ci.mjs), so a second file would add an export line, a `require` and its own coverage to keep up, with no gain.

Why the helpers stay private (agent judgement). Exporting them would add public signatures that later changes must keep (file:ap_portal/CLAUDE.md), and feat.design B4.2 exports only `renderOverdueRisk`. The helpers are covered through `renderOverdueRisk` in the tests.

Kept as named helpers although they could be inlined (agent judgement). `riskClass` and `escapeHtml` could be inlined to save a few lines, but feat.design B4.1 and feat.tasks TSK-07 name them as components. This file follows the approved design rather than trimming it.

3.2 `tests/portal.test.js` *(unit tests)*

Append tests; do not edit the two existing tests. Every error branch needs its own test, because coverage is measured over `src/**` and must stay at 90 or above (file:ap_portal/ci.mjs; feat.tasks section 2, CI bar).

Test (names from feat.design B4.5; `null` and the extra `TypeError` cases from feat.tasks TSK-07)	Covers in `index.js`	Acceptance criteria
`renders overdue risk table`	1.4, 1.8 main path	AC-07.1, AC-07.2
`maps every risk value to its class`	1.3, 1.7	AC-07.3
`returns empty string for no rows` (`[]`, `undefined`, `null`)	1.8 empty branch	AC-07.4
`shows amount strings unchanged` (`"1250.00"`, `"15.00"`, a string with more digits than a JS `Number` holds)	1.6	AC-07.5
`pads short amounts as strings` (`"980.5"` → `980.50`)	1.6 padding	AC-07.6
`rejects non-string amounts`; amount with an exponent, a thousands separator or three decimals → `TypeError`	1.6 error branches	DD-8 (feat.design)
`escapes id and vendor`	1.5	AC-07.7
`unexpected risk renders risk-unknown`	1.7 fallback	AC-07.8
Non-array `rows` → `TypeError`; non-integer `days` → `TypeError`	1.8 error branches	DD-8 (feat.design)
No Risk column, no risk text, no row action in the output	1.8	AC-07.9
Existing tests unchanged; `renderOverdueRisk` exported next to the existing exports	1.9	AC-07.10

Sources: feat.design (B4.5); feat.tasks (TSK-07 "Tests to add"); feat.stories (STORY-07 acceptance criteria).

Minimum-lines approach for the tests (agent judgement): one shared WF-1-style fixture array of CT-1 rows reused across tests, and `assert.throws(..., TypeError)` for the error branches, so each branch is one assertion line. Test data stays synthetic (`SYN-INV-*`, "Synthetic Vendor <letter>") (feat.design SE-4).

3.3 `package.json` *(package metadata)*

Field	Change	Source
`"version"`	Same minor step as `VERSION` in `index.js`; nothing else in the file changes	feat.tasks (section 2, TSK-07 "Done when"); file:ap_portal/package.json

4. Components and layers not changed

Component / layer	Why not	Source
`LABELS`, `formatAmount`, `renderAgingSummary` and their tests	Must not change; the `formatAmount` float issue stays open as TQ-2	feat.tasks (TSK-07); feat.trd (C-9, F-12, TQ-2)
`ci.mjs`	CI already compiles, tests with coverage and runs the static checks the task needs	file:ap_portal/ci.mjs
`README.md`, `CLAUDE.md`	Not in the task's file budget; the README holds only the package description and the CI command	feat.tasks (TSK-07 files in budget); file:ap_portal/README.md; file:ap_portal/CLAUDE.md
Transport or API client between ap_aging and ap_portal	Not added; the consumer parses CT-1 and passes the array. A transport would re-trigger DES-01	feat.design (A3, DD-3, A8)
Host page, navigation, filters, row actions	Out of this design	feat.design (A1 UI boundary); feat.trd (IF-7.8)
Stylesheet for the `risk-*` classes	No stylesheet exists in the repository and none is in the task's file budget; the function only sets the class	tool:repo; feat.tasks (TSK-07 files in budget)
Other repositories (ap_aging, ap_pipeline, ap_forecast) and the end-to-end harness	Other lanes (TSK-01 to TSK-06) and the Testing Agent (TSK-08)	feat.tasks (section 1)
Dependencies	None added: Node.js 22 CommonJS only	feat.design (DD-1)

5. Contract the component consumes (CT-1)

`renderOverdueRisk` is built against the CT-1 field list only, not the ap_aging code, so it can be built in parallel with TSK-06.

Field	Type	How the component uses it	Source
`id`	string	Escaped, first cell	feat.design (A3 CT-1, B4.2)
`vendor`	string	Escaped, second cell	feat.design (A3 CT-1, B4.2)
`days`	integer	Integer text, third cell; `TypeError` if not an integer	feat.design (A3 CT-1, B4.4)
`amount`	two-decimal string	String-only display, `class="amount"` cell	feat.design (A3 CT-1, B4.2); feat.trd (IF-7.5)
`risk`	`high`, `medium`, `low` or `unknown`	Row class only, through the allow-list	feat.design (A3 CT-1, B4.2); feat.trd (IF-7.4)

The input is the array from `JSON.parse` of the ap_aging payload; an empty payload is `[]` (feat.design A3, B4.3).

6. Rules the change must keep

Rule	Source
Keep existing public signatures; add new functions for new behaviour	file:ap_portal/CLAUDE.md; feat.design (header, architecture rule)

Rule	Source
Monetary amounts are never converted to a float: no `Number()`, `parseFloat` or `formatAmount` on the amount	record:memory (D-2026-044, L-011); feat.trd (IF-7.5)
No source line matching `console.log` or `eval(` (comments included); no line longer than 140 characters	file:ap_portal/ci.mjs; feat.tasks (section 2, CI bar)
Coverage of at least 90% with every error branch tested	file:ap_portal/ci.mjs; file:ap_portal/CLAUDE.md; feat.tasks (section 2, CI bar)
Terminology: "Days overdue", never "Late"	feat.prd (NFR-10); feat.design (NF-12)
Commit messages start with the story ID (`STORY-07`); the task ID may follow	feat.tasks (section 2); record:memory (conventions)

7. Pending items that touch these components (carried, non-blocking)

Each is built on its interim design; if an answer differs, only the named item changes.

ID	Affects	Interim design used	Source
TQ-7	1.4, 1.8 header row	Plain `<tr><th>…</th></tr>`, no `<thead>/<tbody>`	feat.trd (IF-7.2, TQ-7); feat.stories (AC-07.2)
TQ-11, OQ-B10	1.6	Pad short amounts as strings	feat.trd (IF-7.5, TQ-11); feat.stories (AC-07.6)
OQ-B8	1.5	Escape `id` and `vendor`	feat.trd (IF-7.7); feat.stories (AC-07.7)
TQ-6	1.3, 1.7	Unknown `risk` renders `risk-unknown`	feat.trd (IF-7.4, TQ-6); feat.stories (AC-07.8)
OQ-B9, PQ-P7	1.8	No Risk column, no risk text, no row actions	feat.trd (IF-7.8); feat.stories (AC-07.9)
DQ-3	1.6, 1.8	Throw `TypeError` on a CT-1 violation	feat.design (DD-8, D3)
TQ-2	`formatAmount` (not changed)	Kept as is; not used by the new code	feat.trd (F-12, TQ-2)

8. Open questions (all non-blocking)

ID	Question	Interim answer	Owner (proposed)
CQ-FE-1	The `risk-*` row classes need styles to be visible to the AP manager, but no stylesheet exists in `ap_portal` and none is in the task's file budget. Who owns the styles?	The function sets the class only; styling stays with the host page, outside this task	UX lead / Developer
CQ-FE-2	No approved Figma frames exist for SC-1 (Figma is not connected); the screens are text frames.	Follow feat.design B4.2 and, in FE-04, feat.screens	UX lead
CQ-FE-3	Exact next version number for `VERSION` and `"version"`	A minor step, confirmed by the lane in FE-04	Developer
RA-1 (carried)	The working branch is `feature/story`, not `feature/STORY-07`, while the convention is `feature/<story>`	Use `feature/story`; commit messages start with `STORY-07`	Developer

Sources: tool:repo; feat.tasks (TSK-07, section 2); feat.stories (STORY-07 frame links, PQ-P6); feat.design (DQ-5); task.repo_access (RA-1); file:ap_portal/CLAUDE.md (branch convention).

9. Validation check

Check	Result	Source
Existing code read before listing changes	Pass: every file in the repository's working tree was read	tool:repo; file:ap_portal/ci.mjs; file:ap_portal/package.json; file:ap_portal/README.md; file:ap_portal/CLAUDE.md
Changes stay inside the TSK-07 file budget	Pass: `src/ap_portal/index.js`, `package.json` (version only), `tests/portal.test.js`	feat.tasks (TSK-07)
Every listed component traces to the design	Pass: items 1.2 to 1.9 map to feat.design B4.1 and B4.2; item 1.1 (header comment) is agent judgement	feat.design (B4)
Every STORY-07 acceptance criterion has a component and a test	Pass: AC-07.1 to AC-07.10 in section 3.2	feat.stories (STORY-07)
No existing signature changed	Pass: only additions plus the `VERSION` value	file:ap_portal/CLAUDE.md; feat.design (B4.1)
Decision log and lessons	Consistent: D-2026-044 and L-011 (no float on amounts)	record:memory
Conflict with an approved output	None	feat.design; feat.tasks; feat.stories
Blocking questions	None	—